

Cadence

A speech synthesis engine that runs anywhere

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Status: In progress — Mo done 3 October 2026; M2 (kernels) next

Roadmap entry: `PROJECTS-ROADMAP.md` #16

Languages: JavaScript (ES2023+) · WebAssembly SIMD · WGSL · PyTorch for training only

1. Summary

Cadence is a text-to-speech and voice-cloning engine in which every compute kernel is written by hand. It runs on the machine in front of the user — faster than real time on an ordinary CPU with no GPU at all, and faster still on the GPU when one is available.

It ships with **a library of more than two hundred built-in male and female voices**, selectable immediately and usable without supplying any audio of your own. Voice cloning from a reference recording is the second, opt-in path, not the primary way the product is used.

There is no API key, no server, no `transformers.js` and no ONNX Runtime. It is free and unlimited for the person using it because the inference is theirs, which is the only honest way anything is ever unlimited and free.

The project is delivered in two phases. **Phase 1** builds the inference engine and ships a working product. **Phase 2** replaces the model weights with weights trained from scratch, ending in a claim very few engineers can make: *I trained the model and I wrote the kernels it runs on.*

2. What this project is for

The existing portfolio says *"I ship business software end to end."* That is valuable and it is crowded. Cadence makes a second, rarer claim: *"I build the layer underneath."*

Almost every developer who works with speech integrates a speech API. Almost nobody implements speech synthesis, and the small number who do are not doing it in a browser tab, on a CPU, in JavaScript. A hand-written SIMD microkernel that turns a neural model into something real time without a GPU is work that machine-learning systems engineers recognise on sight.

It also has a property no other project on the roadmap has: **the output is audible**. A stranger can evaluate it in four seconds by pressing play. Database engines and consensus protocols have to be explained; this one does not.

3. The constraint that defines the project

It must be fast on a CPU. This is not a performance target to be chased after the engine works. It is an architectural decision that has to be made before the first line is written, because it eliminates most of the model space:

Architecture	Why it is ruled out
Diffusion / flow matching	Runs the network 16–32 times per utterance. Hopeless without a GPU.
Autoregressive token models	Latency grows with output length, and the models are large.
Non-autoregressive, single pass	One forward pass. The only shape that reaches real time on a CPU.

The single-pass lineage is therefore the default path. Choose it on day one, or rewrite the engine in month four.

4. The trade-off, stated rather than hidden

Speed, quality and zero-shot voice cloning: **pick two**. The small single-pass models are fast because they are small. The models that convincingly clone a stranger from ten seconds of audio are large and iterative. Nobody has collapsed that, and better kernels will not collapse it either.

So the engine ships two paths behind one interface, and the product is honest about which one the user is on:

	Fast path	Cloning path
Default	Yes	Opt-in
Runs on	CPU, any machine	GPU via WebGPU
Model	Small, non-autoregressive	Larger, cloning-capable
Voices	200+ built-in, male and female	Any voice, from ~10 s of reference audio
Speed	Faster than real time	Slower, still local
Purpose	Makes "unlimited" true everywhere	Makes the feature list complete

One engine, two model sizes, one kernel interface. The interface is the engineering. The honesty about which path is active is the product design.

4.1 The voice library

A large built-in voice set is cheap in exactly the way it looks expensive. In a multi-speaker model a voice is **not a model — it is a speaker embedding**, a single vector of a few hundred floats. Two hundred voices is therefore a few hundred kilobytes of data shipped alongside one model, not two hundred models

to store, load or download. The library costs effectively nothing at runtime, and adding the two-hundred-and-first voice costs nothing either.

Three things supply the voices:

1. **The training corpus.** Open multi-speaker speech corpora contain hundreds to thousands of distinct speakers. A model trained across them yields an embedding per speaker as a by-product.
2. **Sampling the embedding space.** Voices that belong to no real person can be generated by sampling and interpolating within the learned speaker space. This is the same mechanism as voice design, and it means the library is not capped by the corpus. It is also the ethically cleaner source, since a synthetic voice impersonates nobody.
3. **Curation, which is the actual work.** Raw corpus speakers are uneven — noisy recordings, thin coverage, unpleasant timbre. Reaching two hundred *good* voices means auditioning many more, rejecting most, then labelling what survives by gender, age range, accent, and timbre, naming them, and building a picker that makes the set browsable rather than a list of identifiers.

The engineering here is modest; the curation and the labelling are not, and they are scheduled as real work rather than assumed.

4.2 Auditioning a voice before using it

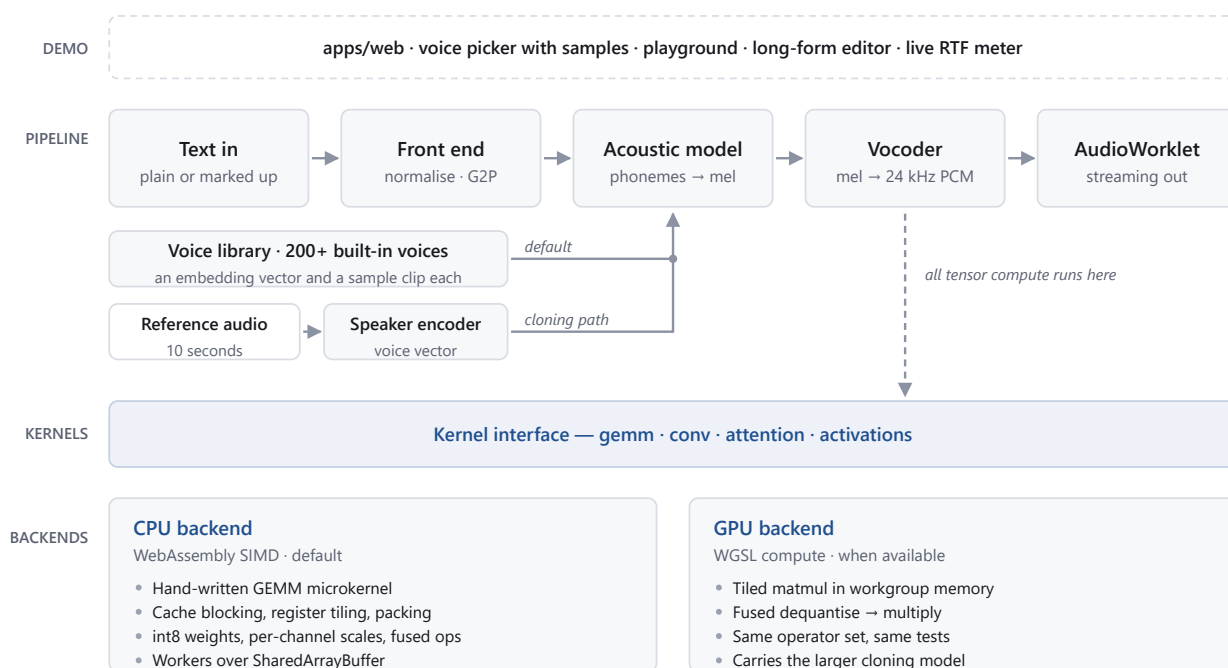
The user journey is **browse, listen, pick, generate**. That requires a short audio sample for every voice in the library, and those samples are pre-rendered rather than synthesised on demand, for two reasons:

- **Scrolling a list of two hundred voices needs instant playback.** Half a second of inference per click is tolerable once and irritating on the twentieth voice.
- **The picker then works before the model has finished loading.** Model download is the slowest part of first use. Pre-rendered samples let someone browse and choose during it, which converts dead waiting time into the most interesting part of the product.

The samples are cheap. Roughly five seconds of Opus per voice is on the order of 15 kB, so the full set lands near 3 MB, lazy-loaded as the list scrolls and cached thereafter. They are generated by the engine itself as the last step of curation: adding a voice generates its sample, so the library and its previews cannot drift apart.

A flat list of two hundred voices is unusable, so the picker is a real deliverable and not a dropdown. It needs search, filtering by gender, age range, accent and language, an inline play control on every row, and a way to keep the handful of voices a given user actually returns to. Custom preview text — hearing a voice say *your* sentence rather than the canned one — is the live-engine path, and the obvious bridge from browsing into generating.

5. Architecture



The same engine package runs unchanged in the browser, in Node and in Electron. No framework is imported below the demo layer.

Both voice sources converge on the same input to the acoustic model: a speaker embedding. Selecting a built-in voice is a lookup in the library; cloning runs the reference audio through the speaker encoder to produce the same kind of vector. Everything downstream is identical, which is why the built-in library costs the engine nothing.

The boundary that matters is the **kernel interface**. Model code is written once against an operator set — `gemm`, `conv`, `attention`, `activations` — and each backend implements that set. The same operator tests run against both, so a disagreement between CPU and GPU output is a test failure rather than a mystery.

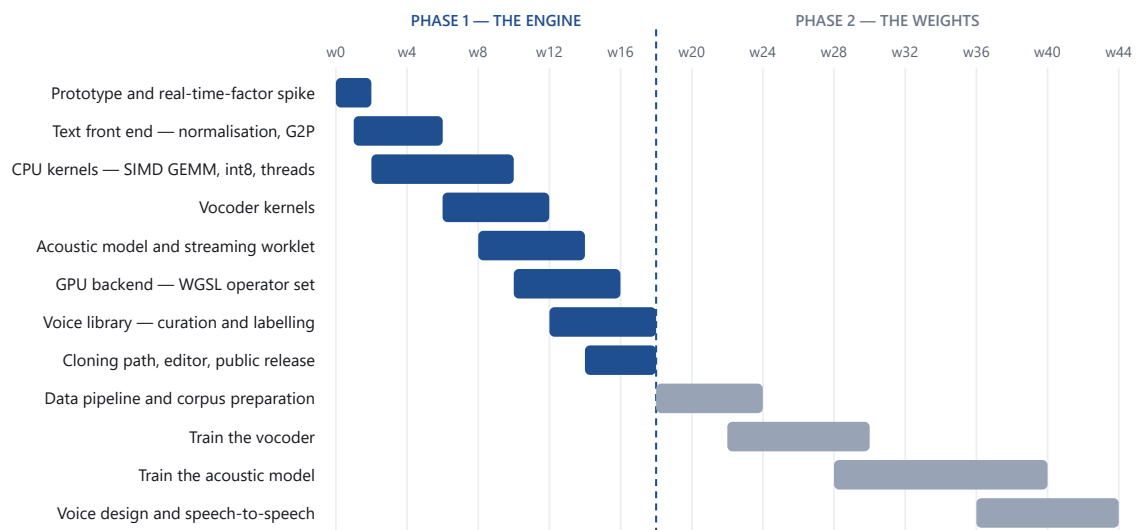
Above it, nothing knows a web framework exists. `packages/cadence` is a plain JavaScript package importable from Node, from a browser and from Electron. `apps/web` is the Next.js demo that makes it visible, and it imports the engine, never the reverse.

6. Scope

The commercial products in this space are not one model; they are roughly seven model families. Attempting all of them in one repository produces an abandoned repository. Synthesis is the coherent half, and it is already a year of work.

Feature	In scope	Where, or why not
Text to speech, multilingual	Yes	Phase 1, both backends
200+ built-in male and female voices	Yes	Phase 1, CPU path — embeddings, not models
A pre-rendered sample for every voice	Yes	Phase 1 — generated by the engine during curation
Searchable, filterable voice picker	Yes	Phase 1, the Next.js demo
Streaming with low time-to-first-audio	Yes	Phase 1
Long-form editor with per-line control	Yes	Phase 1, the Next.js demo
Instant voice cloning from seconds of audio	Yes	Phase 1, GPU path
Professional cloning by fine-tuning	Yes	Phase 2, reuses the training pipeline
Voice design from a text description	Yes	Phase 2, sampling the speaker-embedding space
Speech to speech / voice conversion	Yes	Phase 2, same conditioning, different encoder
Speech to text	No	A separate ASR engine, and its own project
Dubbing	No	Needs ASR, translation and forced alignment on top
Sound-effect and music generation	No	Different model families entirely
Voice isolation and denoise	No	Source separation — belongs with the audio DSP project

7. Plan



Phase 1 — the engine (weeks 0–18)

Validated against existing open weights, used strictly as data and as a correctness reference. Ends in a finished, shippable, offline product.

Milestone	Week	Done means
M0 — the spike	2	A throwaway prototype reports a real-time factor on the target CPU. The number decides the model size before anything else is built.
M1 — it speaks	6	Text normalisation and G2P produce correct phonemes for a held-out test set.
M2 — it is fast	10	The SIMD GEMM microkernel beats the naive implementation by an order of magnitude, with the number published.
M3 — it is audible	14	End-to-end synthesis through the vocoder, streaming to an AudioWorklet with no underruns.
M4 — it is portable	16	The GPU backend passes the same operator tests as the CPU backend.
M4b — it has voices	18	200+ voices auditioned, filtered, labelled by gender, age and accent. Every voice has a generated sample, and the picker makes them searchable and playable before the model loads.
M5 — it ships	18	Cloning path, editor, deployed demo, README, benchmarks, design document.

Phase 2 — the weights (weeks 18–44)

Milestone	Week	Done means
M6 — the corpus	24	A reproducible data pipeline over open speech corpora.
M7 — your vocoder	30	A vocoder trained from scratch, swapped in, with an honest quality comparison.
M8 — your model	40	An acoustic model trained from scratch, and the end-to-end claim becomes true.
M9 — the rest	44	Voice design and speech-to-speech built on the trained embedding space.

The ordering is the risk strategy. Phase 2 needs rented GPUs and can stall on budget; Phase 1 cannot, and leaves a defensible finished project standing either way.

8. How success is measured

Every number below is published in the README, measured on named hardware, including the cases where it loses to the commercial products.

Metric	Target	Why this one
Real-time factor, CPU	< 0.3 on 4 threads, no GPU	The headline claim. Anyone can verify it by running it.
Time to first audio	< 200 ms	What "fast" actually feels like. Throughput matters less than this.
Real-time factor, GPU	Well under CPU, measured	Proves the second backend earns its existence.
Memory ceiling	Runs in a browser tab without crashing it	Quantisation is load-bearing, not decorative.
Voice library size	200+ voices in well under 1 MB	Proves voices are embeddings and not two hundred models.
Preview payload	~3 MB for the full set, lazy-loaded	Browsing must stay cheap, and must work before the model does.
Time to first audition	Playable before the model finishes loading	The picker is the first thing anyone touches.
Kernel speedup	$\geq 10\times$ over the naive baseline	The single most defensible piece of engineering in the repo.
Backend agreement	CPU and GPU within tolerance	Correctness, tested rather than assumed.

9. Risks

Risk	Severity	Response
The CPU is simply not fast enough at the needed quality	High	Mo exists to find this out in week two. If the number is bad, the model size is the dial — learn it in week one, not month four.
Phase 2 compute cost runs away	Medium	Phase 1 ships independently. Train the vocoder first: it is cheap, it is a complete result on its own, and it derisks the pipeline.
Scope creep toward the full product surface	High	Section 6 is the contract. ASR, dubbing and music generation are separate projects or they are nothing.
Weight licensing for the Phase 1 reference models	Medium	Verify terms before use; this corner of the ecosystem shifts. Phase 2 removes the dependency entirely.
Voice cloning misuse	Medium	An inaudible watermark in all output and a consent step on reference upload. Reasoned through openly in the design document.
Silent numerical bugs in hand-written kernels	High	Every operator tested against a reference implementation, both backends, from day one. This is where projects like this actually die.
The 200 voices are uneven, or barely distinguishable from each other	Medium	Curation is scheduled work with a rejection rate, not a data dump. A smaller set of genuinely distinct voices beats a padded count.

10. Deliverables

- `packages/cadence` — the engine. Plain JavaScript, no framework, no bundler assumptions.
- `packages/cadence-kernels` — the CPU and GPU backends behind one operator interface.
- `apps/web` — the Next.js demo: the voice picker, playground, long-form editor, live RTF meter.
- **The voice library as data** — labelled metadata plus a generated sample per voice, built by a repeatable pipeline rather than assembled by hand.
- A **deployed URL** that synthesises speech with no account and no quota.
- A **design document**: the architecture, the trade-offs taken knowingly, what would be done differently.
- **Published benchmarks** on named hardware, including the losses.
- A **written post** on the SIMD microkernel. That post will reach more people than the repository will.

11. Decisions

Taken

1. **Which reference weights for Phase 1** — decided by MO on 3 October 2026, by measurement on the slowest target machine (Intel Core i5-4310U, 2 cores / 4 threads, 8 GB). Every number is in `docs/BENCHMARKS.md`.
 - **Fast path: Piper LibriTTS-R medium.** VITS, about 15M parameters, 904 speakers in one 75 MB model. Natively RTF 0.132 with first audio at 200 ms, inside the 0.3 target, and the best-sounding of the fast candidates in a listening test. Its speakers are the built-in library: a speaker index into one model, as §4.1 predicts.
 - **Higher-quality path: Kokoro-82M, in fp32.** Natively RTF 1.43 here, slower than real time, so it is opt-in rather than the default. Its int8 weights were the slowest format measured (2.87), not the fastest.
 - **Not chosen:** Piper Lessac low and medium (as fast, one voice each) and Piper Lessac high (0.91, too close to real time to stream reliably).

The fast model in the browser (onnxruntime-web, WebAssembly) runs at RTF 0.448 on one thread: faster than real time, 3.4× slower than native, outside the target. WebGPU does not run it (an unsupported integer `GatherND`). ONNX Runtime's profiler puts 82% of the browser's time in convolution, 70% `Conv` and 12% `ConvTranspose`, almost all of it in the vocoder's residual blocks: 32 to 64 channels over 50,000+ samples, kernels of 3 to 7. That is the shape the M2 kernels are written for.

Still open

1. **Languages at launch** — English only is defensible for Phase 1; multilingual widens the G2P work considerably.
 2. **Electron build or browser only** — Electron gives file-system access for long-form work; the browser gives a link anyone can open. The link probably matters more. MO adds a number: on the target machine the native runtime is 3.4× faster than the browser for the same model, until the M2 kernels close that gap.
 3. **Threading fallback** — `SharedArrayBuffer` requires cross-origin isolation headers. A single-threaded path has to exist for environments that will not supply them. On the two-core target, one thread was measured faster than two or four for the fast model, so the fallback may also be the default there.
 4. **How the voice library is sourced** — corpus speakers, sampled synthetic voices, or a mix. Synthetic voices impersonate nobody, which is the cleaner position, but they need the embedding space to be well behaved first. Decide after M3, when there is something to listen to.
 5. **What the preview samples say** — one identical sentence across all voices makes them directly comparable; varied sentences show range but make comparison harder. Probably one shared sentence, with custom preview text available through the live engine.
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Two things to go in with, eyes open

Nobody trains in JavaScript, including you. Phase 2 is PyTorch on a rented box, offline; the exported weights are what the JavaScript engine consumes. The engine is the artifact and the engine is pure JavaScript. Say so in the README rather than implying a pure-JS pipeline that will not exist.

This is not branded as a clone of anything. "Clone of X" reads as derivative. "A speech synthesis engine with hand-written SIMD and GPU kernels" reads as systems work. Same repository, a different category of reader.